

P8139
Homework #1
Due 02/10/26

Show all your steps.

1. A couple who has been married for almost five years has three children. Given that at least two of the children are girls, find the probability that all three children are girls?

Answer:

1/4

2. Three allelic variants, A, B, and C of the red cell acid phosphatase enzyme were found in a sample of 178 people. All genotypes were found in a sample of 178 English people. All genotypes were distinguishable by electrophoresis, and the frequencies in the sample were: genotype AA AB BB AC BC CC frequency (%) 9.6 48.3 34.3 2.8 5.0 0.0. What are the allele frequencies in the sample?

Answer:

$P(A)=35.15\%$, $P(B)=60.95\%$, $P(C)=3.90\%$

3. About 30% of people do not recognize the bitter taste of phenyl-thiocarbamate (PTC). Inability to taste is due to a single autosomal recessive gene. What is the frequency of the non-tasting gene, assuming the population to be in Hardy-Weinberg equilibrium (HWE)?

Answer:

0.55

4. Albinism occurs with a frequency of about 1 in 20000 in European populations. Assuming it to be due to a single autosomal recessive gene, and assuming the population to be in HWE, what is the approximate proportion of people who are carriers?

Answer:

Approx. 0.014

5. The following numbers are the numbers of the human M-N blood groups that were recorded in a sample of American whites: M MN N 1787 3039 1303

a) What the genotype frequencies observed in the sample?

b) What are the gene frequencies?

c) Test whether the genotypes are in HWE.

Answer:

a) MM 29.16%, MN 49.58%, NN 21.26%

b) M: 53.95%, N: 46.05%

c) Do not reject H_0 : HWE